Operating Systems

Tien Pham Van, Dr. rer. nat.

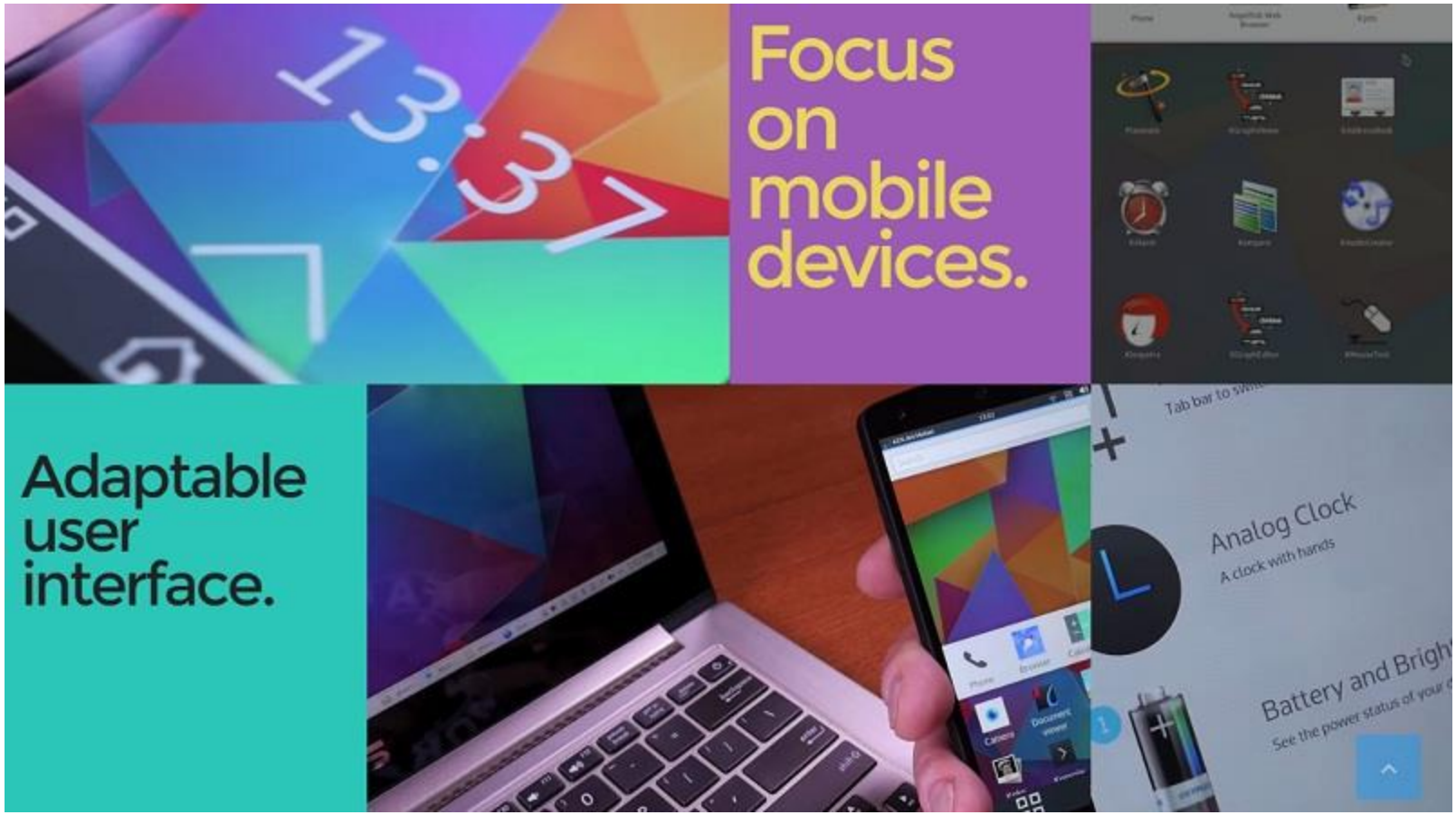
Hanoi University of Science and Technology

Compiled with reference to other presentations

Course Objectives

- To extensively understand concepts and definitions of operating systems
- To be able to work on kernel-native/integrated modules of typical operating systems
- To be able to develop modules/services/drivers
- To be able to approach and investigate advanced topics in operating systems

Where OSs appear…



Course syllabus

- Chapter 1 – Introduction to Operating Systems
- Chapter 2 – Process and Threads
- Chapter 3 – Memory Management
- Chapter 4 – I/O and Drivers
- Chapter 5 – File Systems and Mass Storage
- Chapter 6 – Operating System Security
- Chapter 7 – Virtual Machines
- Chapter 8 – Distributed Systems

References

1. Lecture notes
2. Online reference lectures: video clips
3. Stanford courseware:

   - CS212: https://www.scs.stanford.edu/25sp-cs212/

   - CS240: http://web.stanford.edu/class/cs240/

References

- "*Operating System Concepts, 10th Edition*," Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, 2021.
- *"Modern Operating Systems, 4th Edition (2014),"* Tanenbaum and Bos, Pearson, ISBN 978-0133591620.
- *"Understanding Operating Systems 8th Edition",* Ann McHoes, Ida M. Flynn, Cengage Learning, 2017.

Performance evaluation

- Exercises: 30%
  - short exercises
  - tutorials
- Subproject: 70%